



Jason Pandian
Department of Information Technology
Nehru Institute of Technology
Coimbatore, Tamil Nadu, India
Email: pandiajason@gmail.com

Date: June 2, 2026
Place: Coimbatore

To
Dr. Nicole B. Ellison
Editor-in-Chief
Journal of Computer-Mediated Communication
International Communication Association (Oxford University Press)

Subject: Submission of original research manuscript for consideration

Dear Dr. Ellison,

Please find enclosed our research manuscript titled **“Defying the Speed of Light: How Predictive Temporal AI Architectures Sustain Social Presence and Relational Cohesion in Deep Space Communication,”** which we are submitting for consideration as an original research article for publication in the *Journal of Computer-Mediated Communication*.

This study addresses a critical operational and psychological bottleneck in interplanetary human collaboration: the severe round-trip light-time delay (8 to 45 minutes at Mars distances) that disrupts conversational grounding and isolates crews. Traditional space systems isolate the crew in an “autonomy bubble,” leading to psychological reactance and relational friction with Ground Control. We propose the **Predictive Temporal Bridge (PTB)**, an artificial intelligence-mediated communication (AI-MC) interface that reconstructs a virtual present using dual-AI state integration, thereby sustaining social presence and shared situational awareness (SA).

We evaluate this framework using a high-fidelity interactive simulation of a critical Mars orbital insertion burn anomaly. We trace the subjective and objective effects on communication dynamics using four key indicators: the Operator Presence Fidelity Index (OPFI), the Operator Cognitive Load Index (OCLI), the Effective Command Latency (ECL), and the State Synchronization Accuracy (SSA).

We confirm that:

- This original work has not been published previously or considered elsewhere.
- All authors approved the final manuscript.
- The authors declare no conflicts of competing interest.

Suggested Reviewers:

1. **Dr. Jeffrey T. Hancock** (Stanford University, jeffrey.hancock@stanford.edu): A pioneer in AI-mediated communication (AI-MC) and digital social interaction.
2. **Dr. Joseph B. Walther** (University of California, Santa Barbara, jwalther@comm.ucsb.edu): The originator of the Hyperpersonal Model and a leading expert in computer-mediated social dynamics.
3. **Dr. Gloria R. Leon** (University of Minnesota, leonx002@umn.edu): An expert in the psychological adaptation and communication of astronaut crews in long-duration space analogs.

Generative AI Disclosure: In compliance with the ICA Publications Committee’s Guidelines for the use of generative Artificial Intelligence (AI) for all ICA publications, we disclose that an AI assistant was utilized during the writing and editing process for spelling, grammar verification, structural flow optimization, and formatting the manuscript’s bibliography and layout. No original data analysis, simulation results, or theoretical arguments were generated by the AI tool.

Thank you for your time, effort, and consideration of our work. We look forward to the peer review process.

Sincerely,

A handwritten signature in blue ink, appearing to read 'J. Pandian', is located below the 'Sincerely,' text.

Jason Pandian
Department of Information Technology
Nehru Institute of Technology